Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In S.H.M., the ratio of kinetic energy at mean position to the potential energy when the displacement is half of the amplitude is

(A) $\frac{4}{1}$

(B) $\frac{2}{3}$

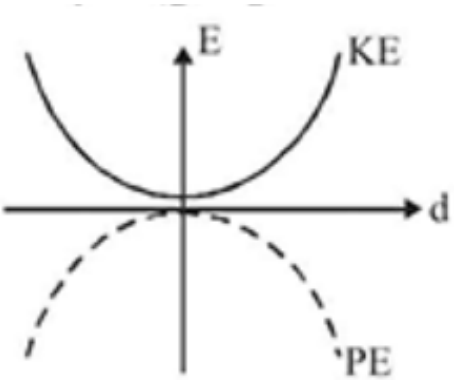
(C) $\frac{4}{3}$

(D) $\frac{1}{2}$

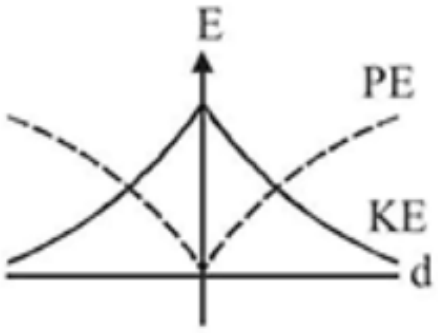
Q.2

For a simple pendulum, a graph is plotted between its kinetic energy (KE) and potential energy (PE) against its displacement d. Which one of the following represents these correctly? (graphs are schematic and not drawn to scale)

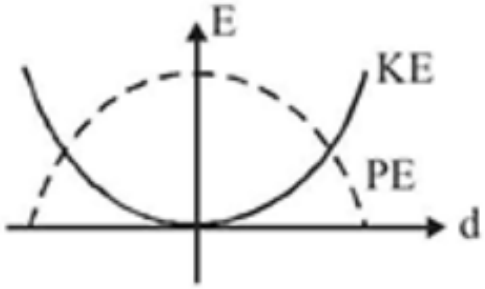
(A)



(B)

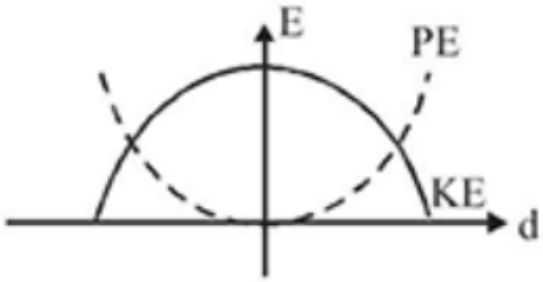


(C)



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(D)



Q.3

The particle executing simple harmonic motion has a kinetic energy $K_0 \cos^2 \omega t$. The maximum values of the potential energy and the total energy are respectively

(A) $K_0/2$ and K_0

(B) K_0 and $2K_0$

(C) K_0 and K_0

(D) 0 and $2K_0$

Q.4

Starting from the origin a body oscillates simple harmonically with a period of 2 s. After what time (in sec) will its kinetic energy be 75% of the total energy?
Answer upto 3 decimal places without rounding off.

(A) 0.156

(B) 0.146

(C) 0.176

(D) 0.166

Q.5

If a is the amplitude of SHM, then $K.E.$ is equal to the P.E. at ..... distance from the mean position.

(A) $\frac{a}{\sqrt{2}}$

(B) $\frac{a}{2}$

(C) $\frac{a}{4}$

(D) a

Q.6

Which of the following is true about total mechanical energy of SHM ?

(A) It is zero at mean position.

(B) It is zero at extreme position.

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(C) It is always zero.

(D) It is never zero.

Q.7

In S.H.M. the ratio of kinetic energy at mean position to the potential energy when the displacement is half of the amplitude is

(A) 4

(B) $\frac{2}{3}$

(C) $\frac{4}{3}$

(D) $\frac{1}{2}$

Q.8

Starting from the origin a body oscillates simple harmonically with a period of 2 s. After what time will its kinetic energy be 75% of the total energy?

(A) $\frac{1}{6}$ s

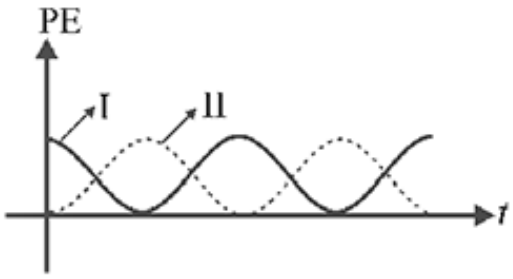
(B) $\frac{1}{4}$ s

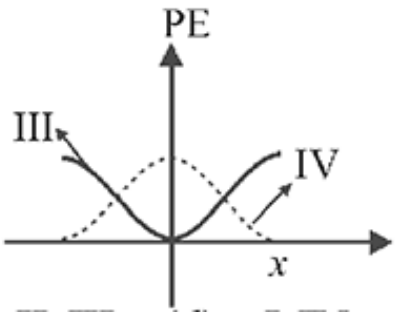
(C) $\frac{1}{3}$ s

(D) $\frac{1}{12}$ s

Q.9

For a particle executing SHM the displacement x is given by $x = A\cos\omega t$. Identify the graph which represents the variation of potential energy (P.E.) as a function of time t and displacement x.





(A) I, III

(B) II, IV

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(C) II, III

(D) I, IV

Q.10

The total energy of the particle executing simple harmonic motion of amplitude A is $100\ \mathrm{J}$. At a distance of 0.707 A from the mean position, its kinetic energy is

(A) $25\ \mathrm{J}$

(B) $50\ \mathrm{J}$

(C) $100\ \mathrm{J}$

(D) $12.5\ \mathrm{J}$

Q.11

When the displacement of a particle executing simple harmonic motion is half of its amplitude, the ratio of its kinetic energy to potential energy is

(A) $1:3$

(B) $2:1$

(C) $3:1$

(D) $1:2$

Q.12

A body executes simple harmonic motion. The potential energy (P.E.), the kinetic energy (K.E.) and total energy (T.E.) are measured as a function of displacement x. Which of the following statement is true?

(A) P.E. is maximum when $x = 0$.

(B) K.E. is maximum when $x = 0$.

(C) T.E. is zero when $x = 0$.

(D) K.E. is maximum when $\mathbf{x}$ is maximum.

Q.13

If $< E >$ and $< U >$ denote the average kinetic and the average potential energies respectively of mass describing a simple harmonic motion, over one period, then the correct relation is

(A) $\langle E \rangle = \langle U \rangle$

(B) $< E >= 2\langle U \rangle$

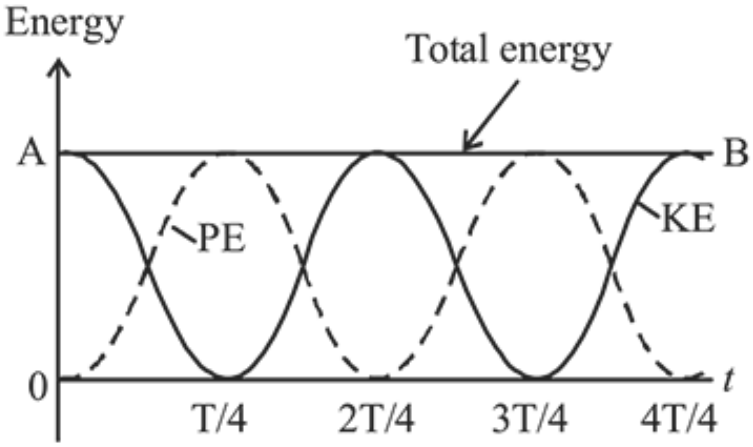
(C) $\langle E \rangle = -2\langle U \rangle$

(D) $\langle E \rangle = -\langle U \rangle$

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Q.14

What do you conclude from the graph about the frequency of KE, PE and SHM?



(A) Frequency of KE and PE is double the frequency of SHM.

(B) Frequency of KE and PE is four times the frequency SHM.

(C) Frequency of PE is double the frequency of K.E.

(D) Frequency of KE and PE is equal to the frequency of SHM.

Q.15

A particle of mass m executes simple harmonic motion with amplitude a and frequency v. The average kinetic energy during its motion from the position of equilibrium to the end is

(A) $2\pi^2 \mathrm{ma}^2\mathrm{v}^2$

(B) $\pi^2 \mathrm{ma}^2\mathrm{v}^2$

(C) $\frac{1}{4}\mathrm{ma}^2\mathrm{v}^2$

(D) $4\pi^2 \mathrm{ma}^2 v^2$

Q.16

A particle of mass $10\mathrm{gm}$ is describing $\mathrm{S.H.M.}$ along a straight line with period of $2\mathrm{sec}$ and amplitude of $10\ \mathrm{cm}$. Its kinetic energy when it is at $5\ \mathrm{cm}$ from its equilibrium position is

(A) $37.5\pi^2$ ergs

(B) $3.75\pi^2 \mathrm{ergs}$

(C) $375\pi^2$ ergs

(D) $0.375\pi^2$ erg

Q.17

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A particle of mass $1\ \mathrm{kg}$ is placed in a potential field. Its potential energy is given by $U = 10x^2 +$ 5. The frequency of oscillations of the particle is given by

(A) $(\sqrt{10})$

(B) $(\sqrt{5})$

(C) $\left(\sqrt{\frac{10}{\pi}}\right)$

(D) $\left(\frac{\sqrt{5}}{\pi}\right)$

Q.18

A body is executing simple harmonic motion. At
a displacement x from mean position, its potential energy is $E_1 = 2\ \mathrm{J}$ and at a displacement y from mean position, its potential energy is $E_2 = 8\ \mathrm{J}$. The potential energy E at a displacement $(x + y)$ from mean position is

(A) $10\ \mathrm{J}$

(B) $14\ \mathrm{J}$

(C) $18\ \mathrm{J}$

(D) $4\ \mathrm{J}$

Q.19

The particle executing simple harmonic motion has a kinetic energy $K_0 \cos^2 \omega t$. The maximum values of the potential energy and the total energy are respectively

(A) $K_0/2$ and K_0

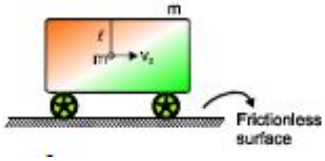
(B) K_0 and $2K_0$

(C) K_0 and K_0

(D) 0 and $2K_0$.

Q.20

A small bob of mass $'m'$ is suspended by a massless string from the ceiling of the cart of the same mass $'m'$ as shown in the figure. The friction between the cart and horizontal ground is negligible. The bob is given a velocity V_0 in the horizontal direction as shown. The maximum height attained by the bob is (initially whole system (bob + string + cart) was at rest).



(A) $\frac{2V_0^2}{g}$

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(B) $\frac{V_0^2}{g}$

(C) $\frac{V_0^2}{4g}$

(D) $\frac{V_0^2}{2g}$

Answers & Solutions

Q.1 Answer:

$\frac{4}{1}$

Solution:

$K.\,\text{E.} = \frac{1}{2}m\omega^2\left(a^2 - y^2\right)$

At mean position, $y = 0$

$\therefore$ **K. E.** $= \frac{1}{2}m\omega^2 a^2$

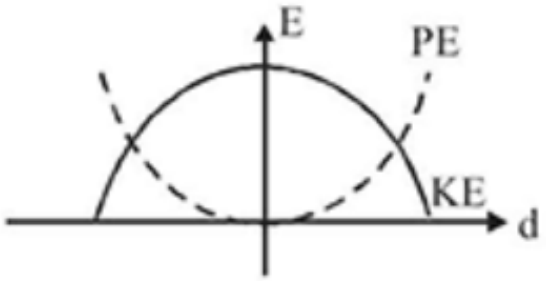
P. E. $= \frac{1}{2}m\omega^2 y^2$

At, $y = \frac{a}{2}$; $P.\,\text{E.} = \frac{1}{2}m\omega^2\frac{a^2}{4} = \frac{1}{8}m\omega^2 a^2$

$\therefore$ **Ratio** $= \frac{(1/2)m\omega^2 a^2}{(1/8)m\omega^2 a^2}$

$= \frac{\text{K.E. at mean position}}{PE \text{ at } a/2} = \frac{4}{1}$

Q.2 Answer:



Solution:

$\text{K.E} = \frac{1}{2}k\left(A^2 - d^2\right)$

and P.E. $= \frac{1}{2}kd^2$

At mean position $d = 0$**. At extrement positions** $d = \text{A}$

Q.3 Answer:

K_0 and K_0

Solution:

We have, $U + K = E$ **where,** $U =$ **potential energy,** $K =$ **Kinetic energy,** $E =$ **Total energy. Also, we know that, in S.H.M., when potential energy is maximum, K.E. is zero and vice-versa.** $\therefore$

$U_{\max} + 0 = E \Rightarrow U_{\max} = E$

Further, $K.E. = \frac{1}{2}m\omega^2 a^2\cos^2\omega t$

But by question, $K.E. = K_0\cos^2\omega t$

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$\therefore K_0 = \frac{1}{2}m\omega^2a^2$

Hence, total energy, $E = \frac{1}{2}m\omega^2a^2 = K_0 \therefore U_{\max} = K_0 \& E = K_0$

Q.4 Answer:

0.166

Solution:

K.E. of a body undergoing SHM is given by, $K.E. = \frac{1}{2}ma^2\omega^2\cos^2\omega t$

$T.E. = \frac{1}{2}ma^2\omega^2$

Given K.E. $= 0.75$ **T.E.**

$\Rightarrow 0.75 = \cos^2\omega t \Rightarrow \omega t = \frac{\pi}{6}$

$\Rightarrow t = \frac{\pi}{6\times\omega} \Rightarrow t = \frac{\pi\times 2}{6\times 2\pi} \Rightarrow t = \frac{1}{6} = 0.1666\text{sec}$

Q.5 Answer:

$\frac{a}{\sqrt{2}}$

Solution:

If displacement of particle is y**, then** $\text{KE} = \frac{1}{2}m\omega^2\left(a^2 - y^2\right) \& \text{P.E.} = \frac{1}{2}m\omega^2y^2$

If $\text{KE} = \text{PE}$ **then** $\frac{1}{2}m\omega^2y^2$

$= \frac{1}{2}m\omega^2a^2 - \frac{1}{2}m\omega^2y^2$

$\Rightarrow 2y^2 = a^2 \quad \therefore y = \frac{a}{\sqrt{2}}$

Q.6 Answer:

It is never zero.

Solution:

Total mechanical energy is constant throughout the motion and equals $\frac{1}{2}m\omega^2A^2$

Q.7 Answer:

4

Solution:

No solution available

Q.8 Answer:

$\frac{1}{6}$ s

Solution:

$K.E. = \frac{1}{2}ma^2\omega^2\cos^2\omega t$

and $T.E. = \frac{1}{2}ma^2\omega^2$

Given K.E. $= 0.75$ **T.E.**

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$\Rightarrow 0.75 = \cos^2 \omega t \Rightarrow \omega t = \frac{\pi}{6}$
$\Rightarrow t = \frac{\pi}{6 \times \omega} \Rightarrow t = \frac{\pi \times 2}{6 \times 2\pi} \Rightarrow t = \frac{1}{6} s$

Q.9 Answer:

I, III

Solution:

In $x = A \cos \omega t$, **the particle starts oscillating from extreme position. So at** $t = 0$, **its potential energy is maximum.**

Q.10 Answer:

50 J

Solution:

No solution available

Q.11 Answer:

$3 : 1$

Solution:

Given, $x = \frac{A}{2}$
$\therefore$ **from** $\mathrm{x} = \mathrm{A} \sin \omega \mathrm{t} \Rightarrow \omega \mathrm{t} = 30^\circ$
And, $\frac{\mathrm{KE}}{\mathrm{PE}} = \cot^2 \omega \mathrm{t} = (\sqrt{3})^2 = 3$

Q.12 Answer:

K.E. is maximum when $x = 0$.

Solution:

$K = \frac{1}{2} m\omega^2 \left(A^2 - x^2\right) \therefore K_{\max} = \frac{1}{2} m\omega^2 A^2,$
at $x = 0$.

Q.13 Answer:

$\langle E \rangle = \langle U \rangle$

Solution:

$E_{av} = U_{av} = \frac{1}{4} m\omega^2 A^2$

Q.14 Answer:

Frequency of KE and PE is double the frequency of SHM.

Solution:

KE **and** PE **completes two vibration in a time during which SHM completes one vibration. Thus frequency of PE or** KE **is double than that of SHM.**

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Q.15 Answer:

$\pi^2 \mathrm{ma}^2 \mathrm{v}^2$

Solution:

The kinetic energy of a particle executing
S.H.M. is given by $\mathrm{K} = \frac{1}{2}\mathrm{ma}^2\omega^2 \sin^2 \omega \mathrm{t}$
$=< \frac{1}{2} m\omega^2 \mathrm{a}^2 \sin^2 \omega \mathrm{t} >= \frac{1}{2}\ \mathrm{m}\omega^2\mathrm{a}^2 < \sin^2 \omega \mathrm{t} >$
$= \frac{1}{2} m\omega^2 a^2 \left(\frac{1}{2}\right)$
$= \frac{1}{4} m\omega^2 \mathrm{a}^2 \quad \left(\because < \sin^2 \theta >= \frac{1}{2}\right)$
$= \frac{1}{4}\mathrm{ma}^2(2\pi v)^2 \quad (\because \omega = 2\pi v)$
or, $\quad < \mathrm{K} >= \pi^2 \mathrm{ma}^2 \mathrm{v}^2$

Q.16 Answer:

$375\pi^2$ ergs

Solution:

Kinetic energy $K = \frac{1}{2} m\omega^2 \left(a^2 - y^2\right)$
$= \frac{1}{2} \times 10 \times \left(\frac{2\pi}{2}\right)^2 \left[10^2 - 5^2\right] = 375\pi^2$ **ergs**

Q.17 Answer:

$\left(\frac{\sqrt{5}}{\pi}\right)$

Solution:

$$F = -\frac{dU}{dx} = -20x$$
$$a = \frac{F}{M} = \frac{-20x}{1} = -20x$$
$$-20x = -\omega^2(x)$$
$$\therefore \omega = \sqrt{20} = 2\sqrt{5}$$
$$f = \frac{\omega}{2\pi} = \frac{2\sqrt{5}}{2\pi} = \left(\frac{\sqrt{5}}{\pi}\right)$$

Q.18 Answer:

18 J

Solution:

$$E_1 = \frac{1}{2}kx^2, E_2 = \frac{1}{2}ky^2,$$
$$E = \frac{1}{2}k(x+y)^2 = \frac{1}{2}kx^2 + \frac{1}{2}ky^2 + kxy$$
$$= E_1 + E_2 + 2\sqrt{E_1 E_2} = 2 + 8 + 2\sqrt{16} = 18 \text{ J}$$

Q.19 Answer:

K_0 and K_0

Solution:

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No solution available

Q.20 Answer:

$\frac{V_0^2}{4g}$

Solution:

A smal bob of .....

By linear momentum conservation in horizontal direction = for (bob + string + cart)

$v = \frac{V_0}{2}$

By mechanical energy conservation for (bob + string + cart + earth)

$\frac{1}{2}mV_a^2 + 0 + 0 = \frac{1}{2}(2m)v^2 + mgh + 0$

$\frac{1}{2}mV_0^2 - \frac{1}{2}(2m)\frac{V_0^2}{4} = mgh$

Solving it. $h = \frac{V_0^2}{4g}$

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